MILITARY STANDARD

INTEROPERABILITY AND PERFORMANCE STANDARDS FOR TACTICAL DIGITAL TRANSMISSION GROUPS (COAXIAL CABLE)



AMSC N/A **AREA TCTS**

DISTRIBUTION STATEMENT A. Approved for public release, distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

FOREWORD

1. This military standard (MIL-STD) is approved and mandatory for use by all Department of Defense (DoD) departments and agencies, in accordance with DoD Directive (DoDD) 4640.11, dated 21 December 1987, Subject: Mandatory Use of Military Telecommunications Standards in the MIL-STD-188 Series.

2. Beneficial comments (recommendations, additions, and deletions) and any pertinent data that may be of use in improving this document should be addressed to:

Director
Joint Tactical Command, Control, and Communications Agency
Washington Office
11440 Isaac Newton Square, North
Reston, VA 22090-5006

by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

3. Originally, MIL-STD-188 covered technical standards for tactical and long-haul communications; later it evolved through revisions (MIL-STD-188A, MIL-STD-188B) into a document applicable to tactical communications only (MIL-STD-188C).

4. The Defense Communications Agency (DCA) published DCA Circulars (DCAC) promulgating standards and engineering criteria applicable to the long-haul Defense Communications System (DCS) and to the technical support of the National Military Command System (NMCS).

5. As a result of a Joint Chiefs of Staff (JCS) action, standards for all military communications are now being published in a MIL-STD-188 series of documents. The MIL-STD-188 series is subdivided into a MIL-STD-188-100 series, covering common standards for tactical and long-haul communications; a MIL-STD-188-200 series, covering standards for tactical communications only; and a MIL-STD-188-300 series, covering standards for long-haul communications only.

6. This interoperability standard is based in part on TRI-TAC Specification TT-B1-2209-0113B, Performance Specification for the Production Family of Digital Group Multiplexers, Pulse Restorers, Cable Driver Modems, and Cable Orderwire Units. In case of a conflict between MIL-STD-188-202 and TT-B1-2209-0113B, the contents of this standard shall be considered a superseding requirement.

7. Where requirements specified in NATO Standard Agreements (STANAG) differ from the requirements as defined in the standard, it will be so identified by placing the word "DIFFERENT" at the end of the appropriate requirements paragraphs of this standard. STANAGs take precedence over this standard, where interoperability is required for combined tactical operations in the European theater, a provision must be made for the equipment to meet the system performance requirements stated in this document and in NATO STANAGS 4206, 4207, and 4210.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

CONTENTS

Paragraph			Page
	1.	SCOPE	1
	1.1	Purpose	1
	1.2	Scope	1
	1.3	Application	1
	1.4	Objectives	1
	2.	REFERENCED DOCUMENTS	3
	2.1	Government documents	3
	2.1.1	Specifications, standards, and handbooks	3
	2.1.2	Other Government publications	3
	2.1.3	Related documents	3
	2.2	Order of precedence	4
	2.3	Source of documents	5
	2.3.1	Government specifications, standards, and handbooks	5
	2.3.2	Other Government publications	5
	3.	DEFINITIONS	
	3.1	Definitions of terms	6
	3.1.1	Asymmetry	6
	3.1.2	Multiplex signal format (MSF)	6
	3.2	Abbreviations and acronyms	6
	4.	GENERAL REQUIREMENTS	8
	4.1	Digital transmission group composition	8
	4.2	Cable interface	8
	4.2.1	Mechanical	8
	4.2.2	CX-11230 connector convention	8
	4.2.3	Cable length	8
	5.	DETAILED REQUIREMENTS	10
	5.1	Traffic line signal	10
	5.1.1	Electrical characteristics	10
	5.1.1.1	Transmitter	10
	5.1.1.1.1	Bit rates	10
	5.1.1.1.2	Modulation	10
	5.1.1.1.3	Level	10
	5.1.1.1.4	Output impedance	10
	5.1.1.1.5	Rise and fall times	11
	5.1.1.1.6	Asymmetry	11
	5.1.1.2	Receiver	11
	5.1.1.2.1	Input impedance	11
	5.1.1.2.2	Performance	11
	5.1.1.2.3	Jitter	11
	5.1.1.2.3.1	Phase	11
	5.1.1.2.3.2	Frequency	11

Source: https://assist.dla.mil -- Downloaded: 2026l0-26T07:28Z
Check the source to verify that this is the currentversion before use.

CONTENTS-Continued

		Page
5.1.2	Multiplex signal formats	12
5.1.2.1	Multiplex frame organization	12
5.1.2.1.1	MSF time slots	12
5.1.2.1.2	Overhead channelization plans	14
5.1.2.1.3	Channel rate	14
5.1.2.2	MSF types	14
5.1.2.2.1	Formatted and unformatted groups	14
5.1.2.2.2	Type 1 group	14
5.1.2.2.3	Type 3 supergroup	15
5.1.2.2.4	Type 4 consolidated group	19
5.1.2.2.4.1	Type 4A consolidated group	19
5.1.2.2.4.2	Type 4B consolidated group	19
5.1.2.2.4.3	Type 4C consolidated group	22
5.1.2.2.4.4	Suppressed frame patterns	22
5.1.3	Multiplex frame synchronization	22
5.1.3.1	Frame control convention	22
5.1.3.2	Frame control operation	22
5.2	Combined digital orderwire	24
5.2.1	Electrical characteristics	24
5.2.1.1	Transmitter	24
5.2.1.1.1	Waveform	24
5.2.1.1.2	Bit rate	24
5.2.1.1.3	Level	24
5.2.1.1.4	Output filtering	24
5.2.1.2	Receiver	24
5.2.2	Format characteristics	24
5.2.2.1	Digital voice orderwire	24
5.2.2.1.1	Digital voice orderwire signaling	24
5.2.2.2	Digital orderwire	24
5.2.2.2.1	Telemetry-combining function	24
5.2.2.2.1.1	Multiplex format	26
5.2.2.2.2	Data channel multiplexer function	26
5.2.2.2.2.1	Multiplex format	26
5.3	Analog maintenance orderwire	27
5.3.1	Electrical characteristics	27
5.3.1.1	Transmitter	27
5.3.1.1.1	Waveform	27
5.3.1.1.2	Signaling	27
5.3.1.1.3	Level	27
5.3.1.2	Receiver	27
6.	NOTES	28
6.1	Subject term (key word) listing	28

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

CONTENTS-Continued

Page

FIGURES

Figure	1.	Designation of coaxial cable connector transmitter/receiver convention	9
	2.	DTG multiplex signal format organization - eight minor frames	13
	3.	Type 3A1 multiplex signal format	17
	4.	Type 3A3 multiplex signal format	18
	5.	Type 4A multiplex signal format	20
	6.	Types 4B and 4C for mixed 32/16 kb/s digital	21
	7.	Resynchronization time-line diagram	23
	8.	Formation of combined digital orderwire	25
	9.	Type 3 group, typical application	33
	10.	Type 4 group, typical application	33

TABLES

Table	I.	Cable length versus bit rate	8
	II.	Traffic line signal group bit rates	10
	III.	Maximum phase jitter versus bit rate	11
	IV.	Overhead channelization plan - eight minor frames	14
	V.	Type 3 group bit rates	16
	VI.	Party-line signaling codes	26
	VII.	Comparision of MIL-STD-188-202 and STANAGs 4206, 4207, 4209, and 4210	35

APPENDIXES

Appendix	A.	List of abbreviations and acronyms used in MIL-STD-188-202	30
	B.	Typical application of Type 3 and Type 4 groups	32
	C.	Military standard and NATO STANAG compatibility	34
	D.	Integrated Services Digital Network (ISDN)	36

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

1. SCOPE

1.1 Purpose. The purpose of this document is to establish technical standards and design objectives (DOs) for the coaxial cable interface of time-division multiplexed (TDM) digital transmission groups (DTGs) for tactical applications.

1.2 Scope. This document standardizes the physical and electrical characteristics and multiplex signal format of the TDM transmission group coaxial cable interface between tactical equipment items.

1.3 Applicability. This document is applicable to the design and development of new equipment, assemblages, and systems that have a TDM transmission group cable interface, and in accordance with direction by the Secretary of Defense, is mandatory for use within the Department of Defense (DoD). It is not intended that existing equipment and systems be immediately converted to comply with the provisions of this standard. New tactical TDM equipment, assemblages, and systems and those undergoing major modification or rehabilitation shall comply with the standards contained in this document, subject to applicable requirements of current procurement regulations. If deviation from this standard is required, then the waiver procedures contained in DoD 4640.11 apply.

1.4 Objectives. The objective of this document is twofold: (1) to provide system performance requirements that ensure interoperation of equipment and systems consistent with Service and Agency requirements, and that achieve the necessary degree of performance and interoperation in the most economical way, and (2) to prevent proliferation of equipment serving the same or a similar function. The variety of equipment shall be the minimum necessary to effectively support the missions of the tactical forces in accordance with Department of Defense Directive (DODD) 4630.5, Compatibility and Commonality of Equipment for Tactical Command, Control, and Communications.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

THIS PAGE INTENTIONALLY LEFT BLANK

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

2. REFERENCED DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this standard to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-C-55583	Cable Assembly, Special Purpose CX-11230( )/G Cable Assembly Adapter CX-10734( )/G
MIL-C-55661	Connector, Plug, Electrical UG-1870( )/U, Connector, Receptacle, Electrical UG-1837( )/U

STANDARDS

FEDERAL

FED-STD-1037	Glossary of Telecommunication Terms.

MILITARY

MIL-STD-188-200	System Design and Engineering Standards for Tactical Communications.

2.1.2 Other Government publications. The following other Government publications form a part of this standard to the extent specified herein.

DoDD 4630.5	Compatibility and Interoperability of Tactical Command, Control, and Communications, and Intelligence Systems.
DoDD 4640.11	Mandatory Use of Military Telecommunications Standards in the MIL-STD-188 Series.

2.1.3 Related documents. The following NATO standards specify system performance requirements that must be met for combined operations in the European theater.

STANAG 4206	The NATO Multi-Channel Tactical Digital Gateway-System Standards.
STANAG 4207	The NATO Multi-Channel Tactical Digital Gateway-Multiplex Group Framing Standards.
STANAG 4210	The NATO Multi-channel Tactical Digital Gateway-Cable Link Standards

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

2.2 Order of precedence. In the event of a conflict between the text of this standard and the references cited herein, the text of this standard shall take precedence. Nothing in this standard, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

2.3 Source of documents.

2.3.1 Government specifications, standards, and handbooks. Copies of the referenced federal and military specifications, standards, and handbooks are available from the Department of Defense Single Stock Point, Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120-5099. For specific acquisition functions, these documents should be obtained from the contracting activity or as directed by the contracting activity.

2.3.2 Other Government publications. Copies of other Government publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

3. DEFINITIONS

3.1 Definitions of terms. Definitions of terms used in this document shall be as specified in FED-STD-1037. Those definitions of terms unique to digital transmission groups and not defined in FED-STD-1037 are provided in the following subparagraphs.

3.1.1 Asymmetry.

$$\text{Asymmetry (in percent)} = \frac{t}{T} \times 100$$

where: T = the bit period.
t = the displacement, in either direction, of the transition in the center of a conditioned diphase zero from the actual center of the bit.

3.1.2 Multiplex signal format (MSF). An organization of overhead and information channel time slots with respect to the framing subchannel, as further defined in 5.1.2 herein.

3.2 Abbreviations and acronyms. Abbreviations and acronyms used in this document are defined in appendix A.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

THIS PAGE INTENTIONALLY LEFT BLANK

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

4. GENERAL REQUIREMENTS

This standard defines the interface requirements for digital transmission groups. The interface is defined at the shelter wall, and the interfacing elements are the shelter and CX-11230 coaxial cable.

Note: The CX-11230 coaxial cable and associated UG-1870 connector plug are part of the standard interconnecting cable assembly used by all U.S. Services for tactical communications, as specified in MIL-C-55583 and MIL-C-55661. (STANAG 4210, Cables and Connectors, para. 4 - DIFFERENT)

4.1 Digital transmission group composition. A digital transmission group (DTG) shall consist of a conditioned diphase modulated TDM traffic line signal and an orderwire signal. The orderwire signal will consist of either a 32-kb/s combined digital orderwire (CDOW) or an analog voice orderwire (AVOW) signal. (STANAG 4206, Engineering Orderwire, para. 24 - DIFFERENT)

4.2 Cable interface.

4.2.1 Mechanical. The CX-11230 cable carrying the time-division multiplexed group shall be terminated by a UG-1837 bulkhead connector receptacle. (STANAG 4210, Cables and Connectors, para. 4 - DIFFERENT)

4.2.2 CX-11230 connector convention. The female receptacle (socket) shall be used for the transmit signal leaving the shelter wall, as shown in figure 1. (STANAG 4210, Cables and Connectors, para. 4 - DIFFERENT)

4.2.3 Cable length. The cable length is dependent upon the traffic line signal bit rate, as listed in table I. (STANAG 4210, Interconnection Point, para. 1f - DIFFERENT)

TABLE I. Cable length versus bit rate.

Bit rate, kb/s	Length, km
128/256/512	0 to 3.2
2048	0 to 1.6

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

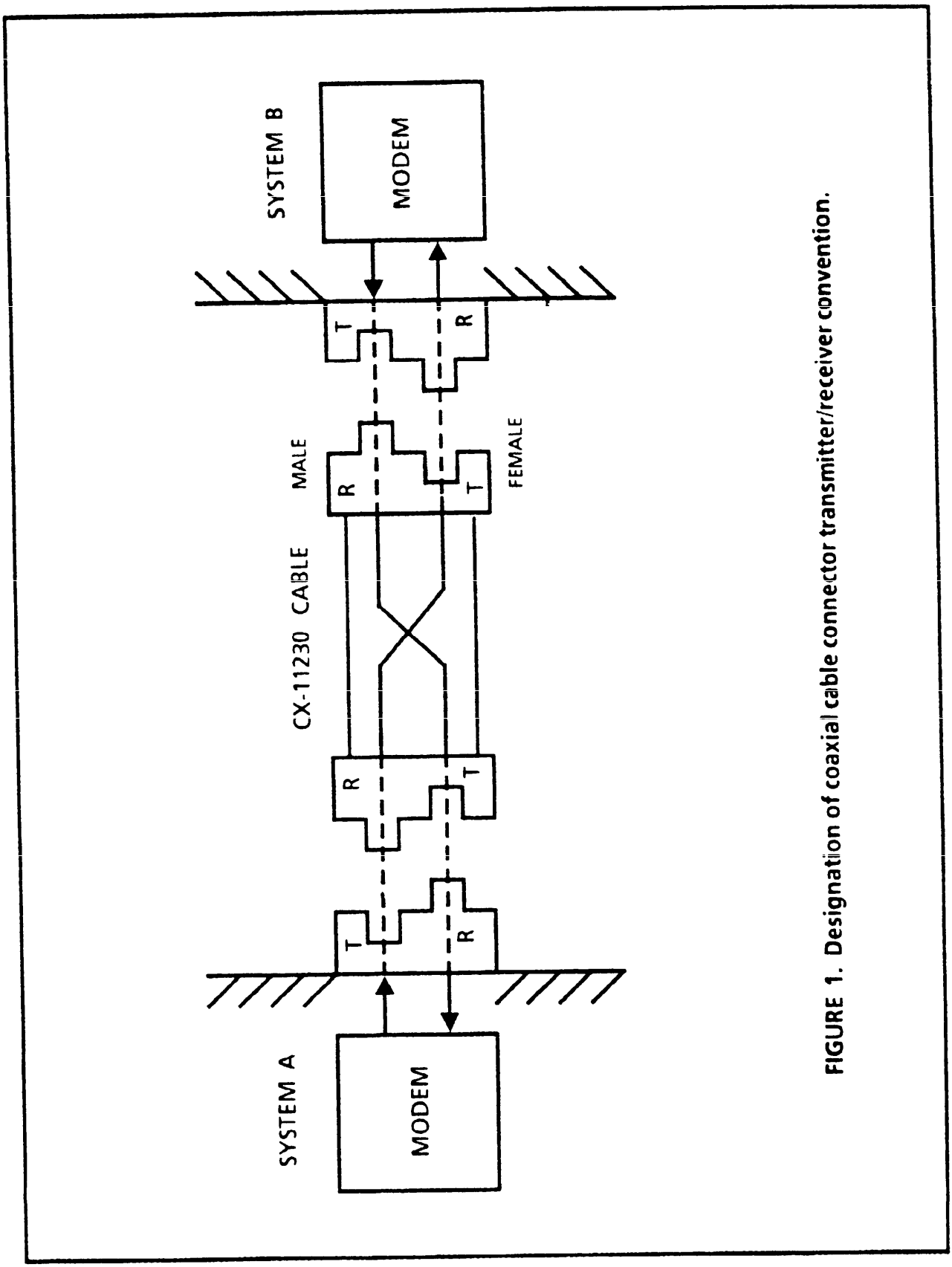


FIGURE 1. Designation of coaxial cable connector transmitter/receiver convention.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

5. DETAILED REQUIREMENTS

5.1 Traffic line signal.

5.1.1 Electrical characteristics.

5.1.1.1 Transmitter.

5.1.1.1.1 Bit rates. The traffic line signal group bit rates are defined in table II.

TABLE II. Traffic line signal group bit rates.

Total Channels (Includes Primary Overhead)	Maximum Available Traffic Channels	Bit Rate for 16-kb/s Channel (kb/s)	Bit Rate for 32-kb/s Channel (kb/s)
8	7	128	256
*16	15	256	512
32	31	512	--
64	63	--	2048
128	127	2048	--

NOTE: Tactical equipment developed prior to this standard employed additional bit rates. This standard specifies a minimum number of bit rates that will provide the desired interoperability in future tactical systems. *(STANAG 4207 - Table 2, maximum available traffic channels 14 - DIFFERENT)

5.1.1.1.2 Modulation. The traffic line signal group bit rates shall employ conditioned diphase modulation, as described in the MIL-STD-188-200 subparagraph entitled "Conditioned diphase modulation."

5.1.1.1.3 Level. The amplitude of the transmit traffic line signal shall be 3 volts peak-to-peak (Vp-p) plus or minus 20 percent, as measured at the transmit terminals, into a 58-ohm dummy load. (STANAG 4210 Interconnection Points, para. 1c - DIFFERENT)

5.1.1.1.4 Output impedance. The output impedance shall be 50 to 65 ohms from 400 Hz to 1.0 MHz and 55 to 60 ohms from 1.0 MHz to 4.0 MHz, unbalanced. (STANAG 4210 Interconnection Point , para. 1a. thru f. - DIFFERENT)

Source: https://assist.dla.mil -- Downloaded: 2026l0-26T07:28Z
Check the source to verify that this is the currentversion before use.

5.1.1.1.5 Rise and fall times. The rise and fall times of the traffic line signal shall be no greater than 15 percent of the pulsewidth of the highest applicable bit rate.

5.1.1.1.6 Asymmetry. The asymmetry measured at the transmit terminals shall be no greater than 5 percent.

5.1.1.2 Receiver.

5.1.1.2.1 Input impedance. The input impedance shall be 50 to 65 ohms from 400 Hz to 1.0 MHz and 55 to 60 ohms from 1.0 MHz to 4.0 MHz, unbalanced. (STANAG 4210, para. 1a. thru f. - DIFFERENT)

5.1.1.2.2 Performance. The receiver shall operate with a bit error ratio (BER) of better than 10^{-9} under any combination of the following conditions:

a. At any specified bit rate.

b. With or without an orderwire (AVOW or CDOW) signal present.

c. With maximum transmitter asymmetry.

d. At maximum and minimum transmitter levels.

e. With any organization of traffic bits.

f. For any specified cable length.

5.1.1.2.3 Jitter. (STANAG 4206, Timing and Jitter, para. 12 thru 15 - DIFFERENT)

5.1.1.2.3.1 Phase. The receiver shall accommodate a P-P jitter no greater than the values listed in table III for the given bit rate. Values provided are the maximum accumulated phase differences in bits between output group signals and the corresponding input signals for the group rates listed.

TABLE III. Maximum phase jitter versus bit rate.

Bit Rate, kb/s	Peak-to-peak Jitter, bits
128/256	2.5
512	2.6
2048	3.7

5.1.1.2.3.2 Frequency. The maximum frequency deviation of any output group signal as measured from the nominal frequency of the input group signal shall be ± 1.25 parts in 10^6 or less for long term group frequency offsets within ± 1 part in 10^6 of nominal, and ± 2 parts in 10^6 or less for long term offsets within ± 30 parts in 10^6 of nominal. This deviation shall not be exceeded for up to three tandem MGM groups. In all cases, the long term average output frequency shall equal the long term average input frequency.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

5.1.2 Multiplex signal formats. The traffic line signal shall be made up of overhead channel bits and information bits. These bits shall be organized as defined herein. (STANAG 4206, Multiplex Group-Frame Structure, para. 6 thru 11, and 4207, Multiplex Frame Organization, para. 1 thru 5 - DIFFERENT for Section)

5.1.2.1 Multiplex frame organization. Figure 2 depicts a representative multiplex signal format (MSF). The MSF has one major frame that has a time period of 0.5 milliseconds for a 16-kb/s channel rate (see 5.1.2.1.3) or 0.25 milliseconds for a 32 kb/s channel rate. Each major frame is organized into eight minor frames, and each minor frame has an overhead subchannel time slot and some number of information bit time slots. The eight overhead subchannels collectively make up the primary overhead channel. (STANAG 4206, Multiplex Group-Frame Structure, para. 6 thru 11 - DIFFERENT)

The first bit in each major frame is defined as the framing bit, and the sequence of framing bits from each major frame is designated as the framing subchannel. From one major frame to the next, this frame bit produces the frame patterns described in 5.1.3. Multiplexing and demultiplexing of the bits contained in all other time slots shall use a fixed relationship between a time slot assignment and the postion of the time slot with respect to the framing bit. (STANAG 4207, Multiplex Frame Organization, para. 1 thru 5 - DIFFERENT)

5.1.2.1.1 MSF time slots. The following terms are defined with respect to MSF time slots:

a. Primary overhead time slots. The primary overhead time slot structure shall consist of the framing bit time slot and seven time slots (see figure 2) that may be used to provide up to seven subchannels.

b. Information channel time slots. Information channel time slots shall include secondary overhead time slots (see 5.1.2.2.4) and traffic channel time slots.

c. Channelization plan. The allocation of particular time slots within a major frame to specific channels or subchannels.

d. Time slot address. A particular time slot or set of time slots is referenced to the frame time slot by specifying the minor frame number (0 through 7) and traffic time slot number (0 through N) -- a rectangular coordinate representation.

e Frame alignment. The invariant allocation of secondary overhead subchannels with the suppressed frame pattern (1100) having an address of minor frame 0 and a traffic time slot number less than or equal to (N + 1)/2.

f. Subchannel. A contiguous bit stream placed in or extracted from a particular time slot or set of time slots associated with a primary or secondary overhead channel.

g. Suppressed frame pattern. A logical 1100 sequence assigned to the first subchannel in a secondary overhead channel.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

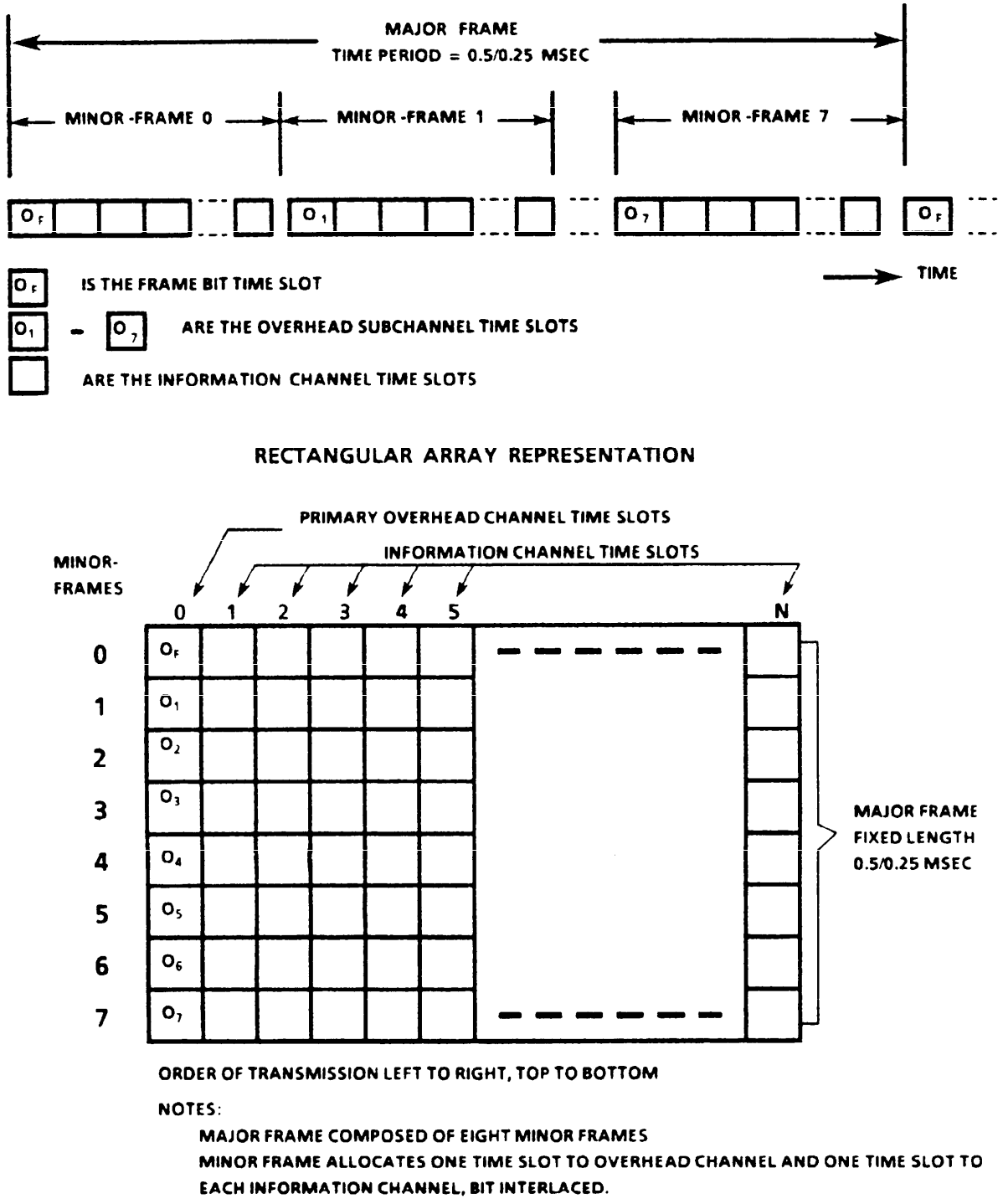


FIGURE 2. DTG multiplex signal format organization - eight minor frames.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

5.1.2.1.2 Overhead channelization plans. There are three primary overhead channelization plans, designated as Type L, Type O, and Type S. Reference table IV for the overhead channelization plan.

TABLE IV. Overhead channelization plan - eight minor frames.

TIME SLOT DESIGNATION	OVERHEAD SUBCHANNEL ASSIGNMENT		
	TYPE L	TYPE O	TYPE S
O_F	FRAMING	FRAMING	FRAMING
O_1	NOT USED	GENERAL USE	COMMON SIGNALING SUBCHANNEL
O_2	NOT USED	GENERAL USE	COMMON SIGNALING SUBCHANNEL
O_3	NOT USED	GENERAL USE	COMMON SIGNALING SUBCHANNEL
O_4	NOT USED	GENERAL USE	COMMON SIGNALING SUBCHANNEL
O_5	TELEMETRY	GENERAL USE	SYSTEM CONTROL SUBCHANNEL
O_6	TELEMETRY	GENERAL USE	SYSTEM CONTROL SUBCHANNEL
O_7	TELEMETRY	GENERAL USE	SYSTEM CONTROL SUBCHANNEL

5.1.2.1.3 Channel rate. Within the MSF, all channels, primary overhead and information, shall operate at 16-kb/s and 32-kb/s channel rates as specified herein. The framing subchannel rate shall be 2 kb/s for a 16-kb/s channel rate and 4 kb/s for a 32-kb/s channel rate. (STANAG 4206, Multiplex Group Frame Structure, para. 6 thru 11 - DIFFERENT)

5.1.2.2 MSF types.

5.1.2.2.1 Formatted and unformatted groups.

a. Formatted group - A formatted group has characteristics as specified in 5.1.2.2.2 through 5.1.2.2.4.

b. Unformatted group - An unformatted group has a bit rate as specified herein. The MSF requirements do not apply to an unformatted group.

c. Supergroup and combined group - A supergroup or combined group is made up of a formatted group plus any combination of formatted and unformatted groups.

5.1.2.2.2 Type 1 group.

The Type 1 group shall use an eight minor frame MSF and is depicted in figure 2. The Type 1 group shall have a time slot structure with one primary overhead channel and N information channels, where N = 7, 15, 31, 63, or 127. The following rules apply to the formation and use of this MSF:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

a. All channels shall be either 16-kb/s or 32-kb/s.

b. An L-,O-, or S-type overhead plan shall be used.

c. All unused time slots in the overhead channel shall be set to one.

d. Information channel time slots that are not in use (i.e., idle channels) shall be set to one.

e. All information channels shall be traffic channels.

5.1.2.2.3 Type 3 supergroup. See appendix B for a typical application. The Type 3 supergroup shall have a time slot structure that will combine one Type 1 group with from one to three other groups. The other groups can be any combination of Type 1 groups or unformatted groups. In the Type 3 supergroup, one Type 1 group shall be designated the Type 3 supergroup, one Type 1 group shall be designated the primary group, and all other groups shall be designated secondary groups. The overhead channel in a secondary group derived from a Type 1 group is designated a secondary overhead channel. The Type 3 supergroup shall have a time slot structure with one primary overhead channel and N information channels, where N = 7, 15, 31, 63, or 127. The following rules apply to the formation and use of this supergroup:

a. All channels shall be either 16-kb/s or 32-kb/s.

b. The primary overhead time slot structure from one Type 1 group shall be selected to provide the primary overhead function for the Type 3 supergroup.

c. All characteristics of Type 1 groups shall be preserved, except as specified otherwise in e below.

d. The four group input rates and the Type 3 supergroup rate shall be any combination, as specified in table V. The frame formats are defined in figures 3 and 4.

e. For all secondary groups made up of Type 1 groups, the framing bit time slot shall contain a suppressed frame pattern that is defined as 1100. A suppressed frame pattern time slot shall be assigned to the first minor frame, and the remaining secondary group overhead subchannel slots shall be distributed in the prescribed order in minor frames one through seven.

f. The group inputs shall be combined, as shown in figures 3 and 4.

(1) When all secondary group inputs are Type 1 groups, the structure shall be as shown in figures 3 and 4.

(2) When one or more secondary groups are unformatted, the time slots that are shown as being used for a Type 1 group shall be used to provide the time slots for an unformatted group.

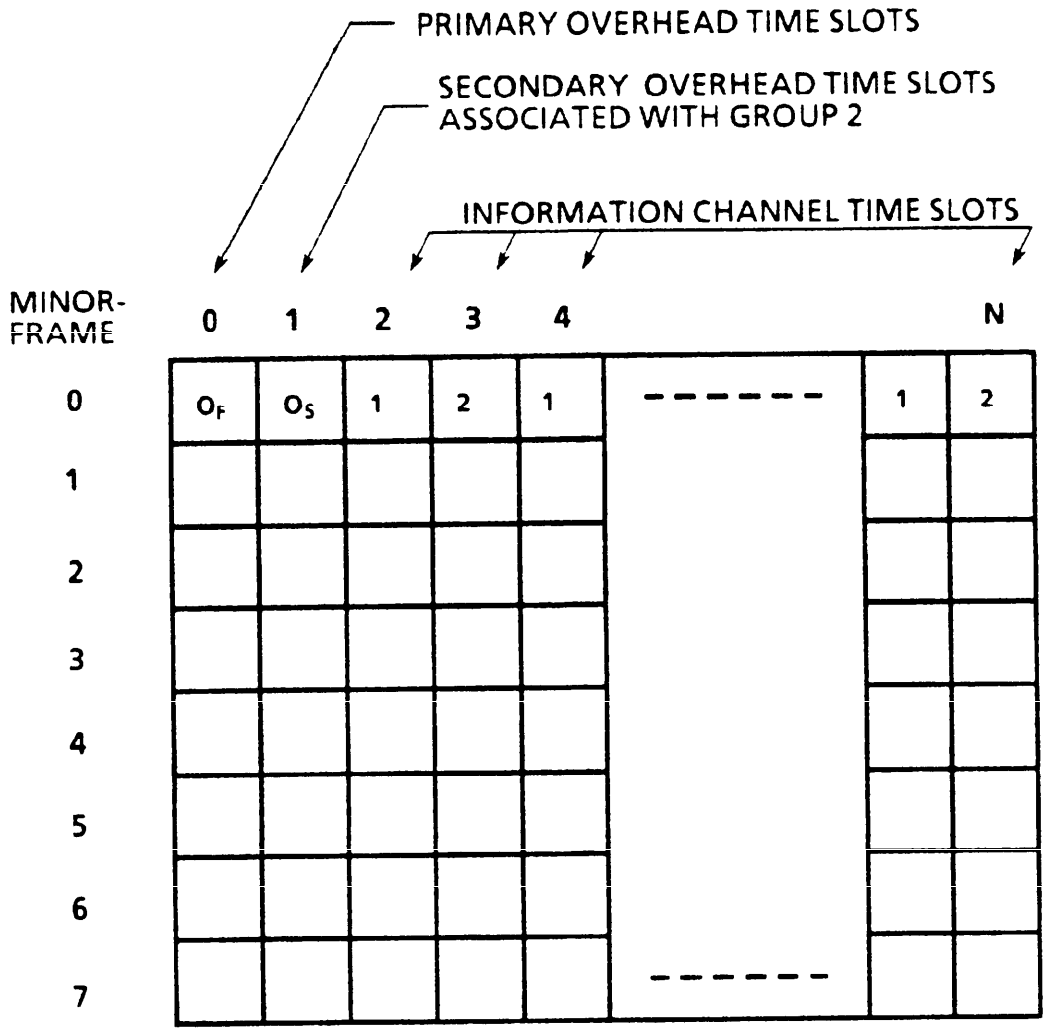
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

TABLE V. Type 3 group bit rates.

INPUT GROUP BIT RATES (kb/s)				TYPE 3 BIT RATE (kb/s)	FRAME FORMAT
INPUT #1	INPUT #2	INPUT #3	INPUT #4		
512	512	128/256/512	0/128/256/512	2048	A3
512	256	512	128/256	2048	A3
256	256	0	0	512	A1
128	128	0	0	256	A1
128	128	128	0/128	512	A3

NOTE: Some rate combinations do not utilize the MSF capacity associated with the combined rate entries. In these cases, the lower individual group rates shall be multiple-sampled to fill out the available frame time slots.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

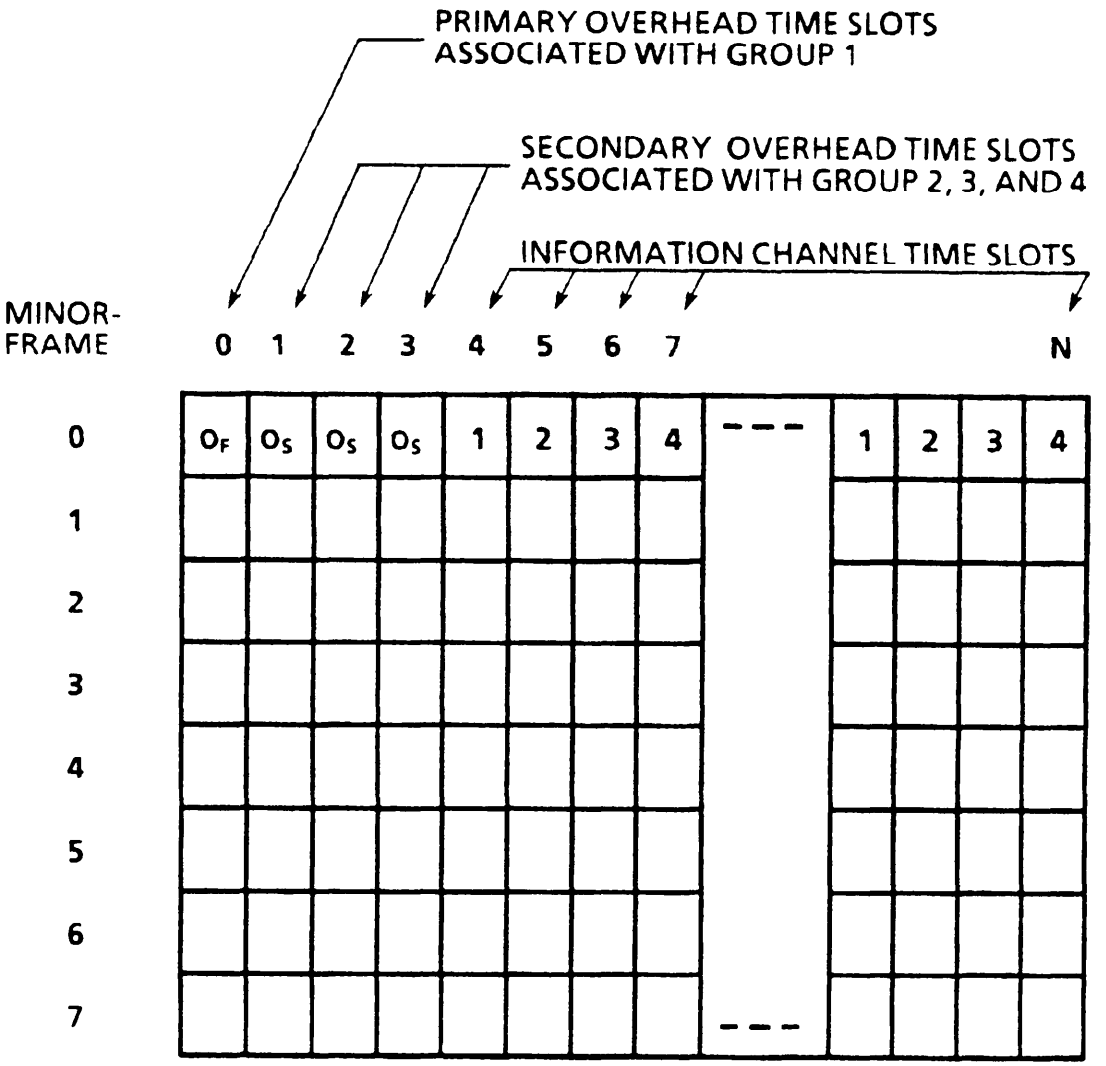


NOTES:

O_F - FRAME BIT TIME SLOT
O_S - SUPPRESSED FRAME TIME SLOT
1 - INFORMATION CHANNEL ASSOCIATED WITH PRIMARY OVERHEAD - GROUP 1
2 - INFORMATION CHANNEL ASSOCIATED WITH SECONDARY OVERHEAD - GROUP 2

FIGURE 3. Type 3A1 multiplex signal format.

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:28Z
Check the source to verify that this is the current version before use.



NOTES:

O_F - FRAME BIT TIME SLOT
O_S - SUPPRESSED FRAME TIME SLOT
1 - INFORMATION CHANNEL ASSOCIATED WITH PRIMARY OVERHEAD - GROUP 1
2 - INFORMATION CHANNEL ASSOCIATED WITH SECONDARY OVERHEAD - GROUP 2
3 - INFORMATION CHANNEL ASSOCIATED WITH SECONDARY OVERHEAD - GROUP 3
4 - INFORMATION CHANNEL ASSOCIATED WITH SECONDARY OVERHEAD - GROUP 4

FIGURE 4. Type 3A3 multiplex signal format.

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:28Z
Check the source to verify that this is the current version before use.

5.1.2.2.4 Type 4 consolidated group. See appendix B for a typical application. The Type 4 consolidated group shall utilize an eight minor frame MSF. The Type 4 consolidated group shall have a time slot structure that combines one Type 1 group with any combination of Type 1 groups or unformatted groups. In the Type 4 consolidated group, one Type 1 group is designated as the primary group and all other groups are designated secondary groups. The following shall apply to the Type 4 consolidated group:

a. Secondary overhead plans must be positioned within the first (N + 1)/2 time slots (slots are numbered in increasing numeric order beginning with the primary overhead slot as number zero).

b. The traffic time slots associated with the primary and secondary overhead time slots may be positioned anywhere within the frame.

c. Groups derived from Type 1 groups using L-Type overhead plans may be combined without including their associated L-Type overhead channels.

5.1.2.2.4.1 Type 4A consolidated group. The Type 4A consolidated group shall be formed by combining only 32-kb/s information channels and overhead channels, or only 16-kb/s information channels and overhead channels. The consolidated group shall be structured in accordance with figure 5.

5.1.2.2.4.2 Type 4B consolidated group. The Type 4B consolidated group shall be formed by combining 32-kb/s information and overhead channels with 16-kb/s traffic and overhead channels. The consolidated group shall be structured in accordance with figure 6. The following shall apply to the Type 4B consolidated group:

a. A given bit of a 16-kb/s information channel shall appear twice; that is, in two successive minor frames in the same information time-slot position.

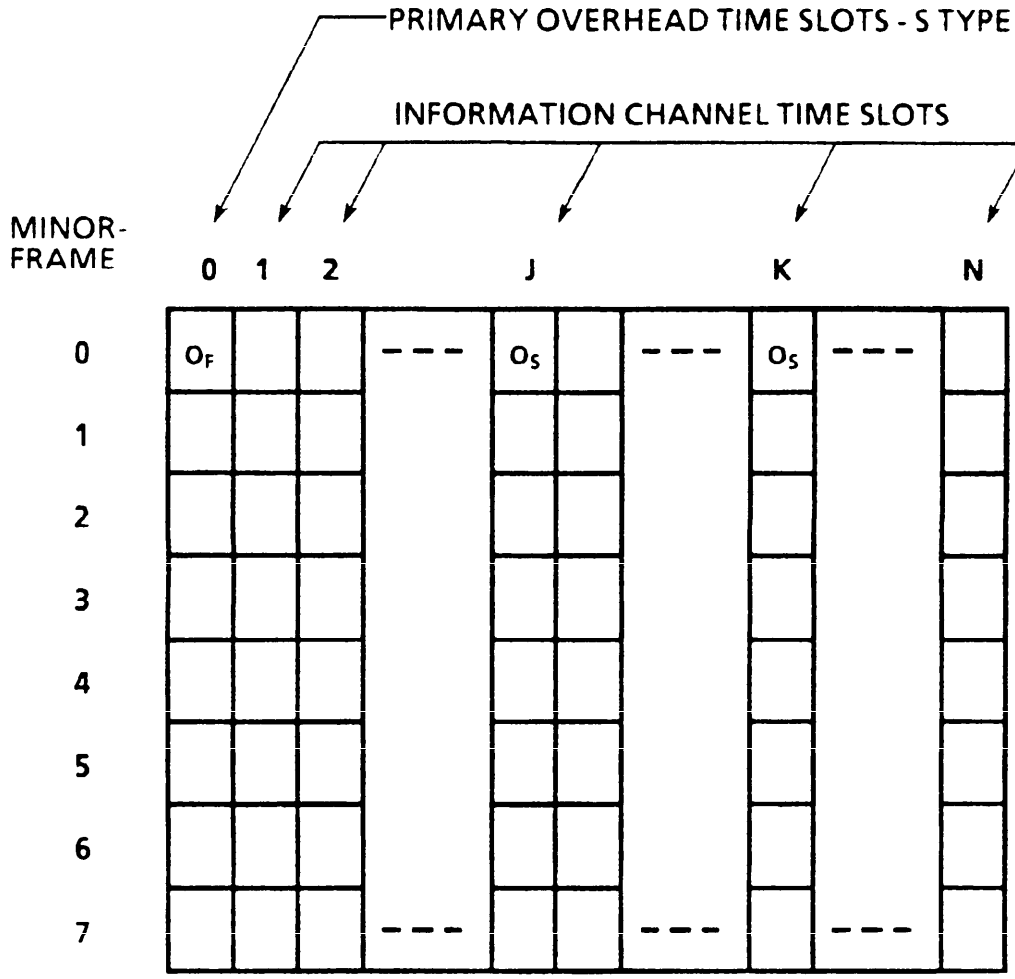
b. A 16-kb/s secondary overhead channel shall appear twice; that is, in two successive major frames in the same information time-slot position.

c. A 16-kb/s secondary overhead channel shall span two major frames of the Type 4B consolidated group as follows: the suppressed frame pattern (1100) shall be placed in the O_S positions of the secondary overhead time slots. The suppressed frame pattern shall be identical in rate and appearance to that of a secondary 32-kb/s overhead channel. All subchannels of a 16-kb/s overhead channel shall appear in the same order and relative position as those of a 32-kb/s overhead channel in a given major frame. Each bit shall appear twice in two successive major frames in the same relative time-slot position.

d. The maximum number of 16-kb/s groups that may be combined with 32-kb/s groups shall be four.

e. The 16-kb/s groups shall be Type 1.

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:28Z
Check the source to verify that this is the current version before use.



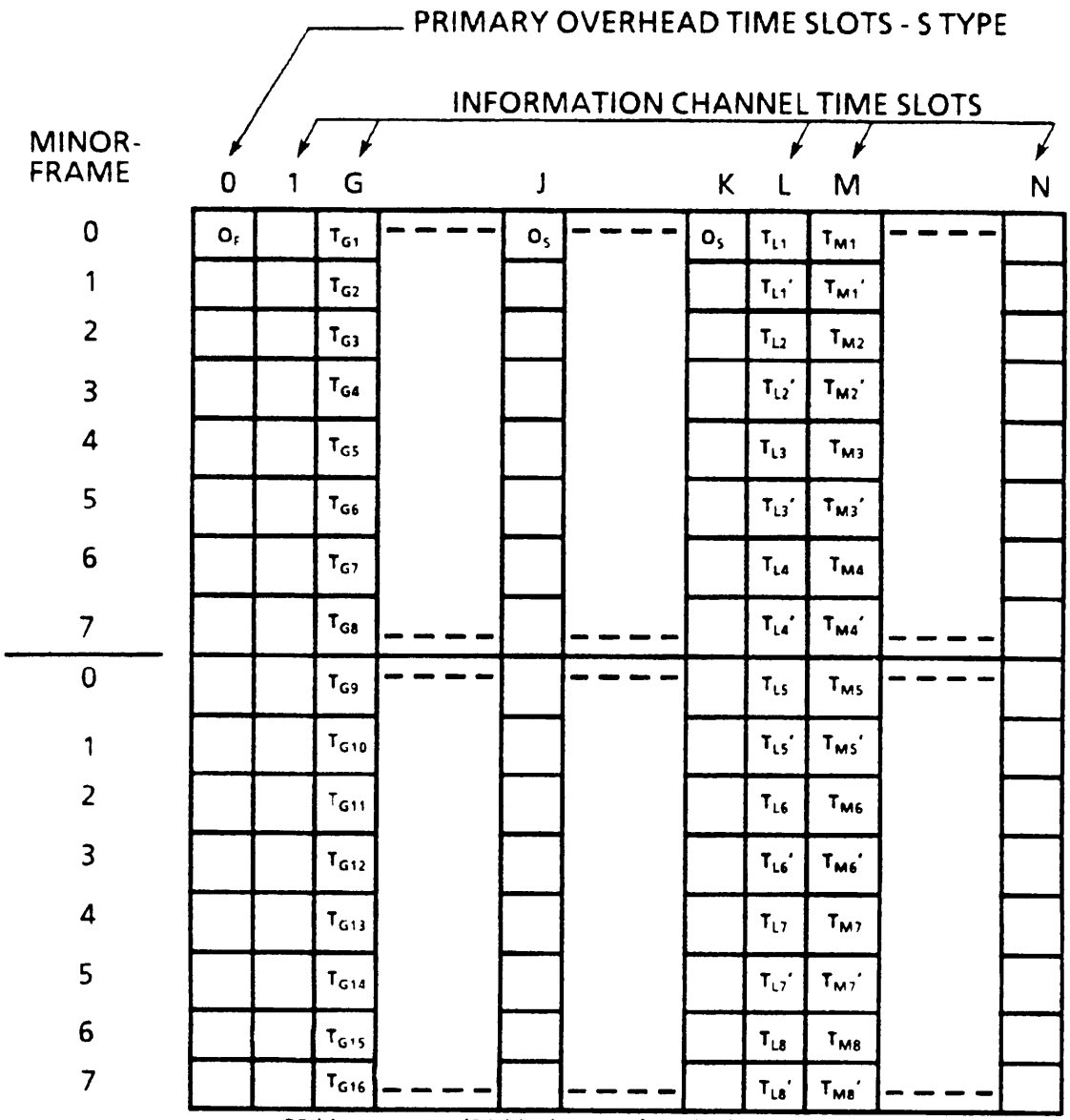
NOTES:

$K \leqq \frac{N+1}{2}$

O_F - FRAMING BIT TIME SLOT
O_S - SUPPRESSED FRAME TIME SLOT

FIGURE 5. Type 4A multiplex signal format.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.



NOTES:

$K \leq \frac{N + 1}{2}$

O_F - FRAME BIT TIME SLOT

O_S - SUPPRESSED FRAME TIME SLOT

T_1-T_8, T_1'-T_8' - 16 kb/s - INFORMATION AND REDUNDANT INFORMATION TIME SLOTS (TL1 = TL1', ETC.)

FIGURE 6. Types 4B and 4C for mixed 32/16-kb/s digital transmission groups.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

5.1.2.2.4.3 Type 4C consolidated group. The Type 4C consolidated group shall be the same as Type 4B with no restrictions in the number of 16-kb/s groups. It shall also include Type 1 groups and Type 3 supergroups.

5.1.2.2.4.4 Suppressed frame patterns. In all Type 4 consolidated groups, suppressed frame patterns shall be assigned to the first minor frame. The remaining secondary overhead channel slots shall be distributed in the prescribed order in minor frames one through seven.

5.1.3 Multiplex frame synchronization.

5.1.3.1 Frame control convention. The following shall comprise the frame control convention:

a. The Frame Request (FR) signal consists of a major frame in which the framing-bit time slot is set to one and all other major frame time slots are set to zero. The all-ones pattern in the framing-bit time slot is called the out-of-sync frame pattern.

b. The Frame Request Acknowledge (FRA) signal is a major frame in which the framing-bit time slot is set to produce an alternating one-zero pattern, called the in-sync frame pattern. All other major frame slots are set to zero.

5.1.3.2 Frame control operation. Frame control operations shall be in accordance with figure 7 and the following:

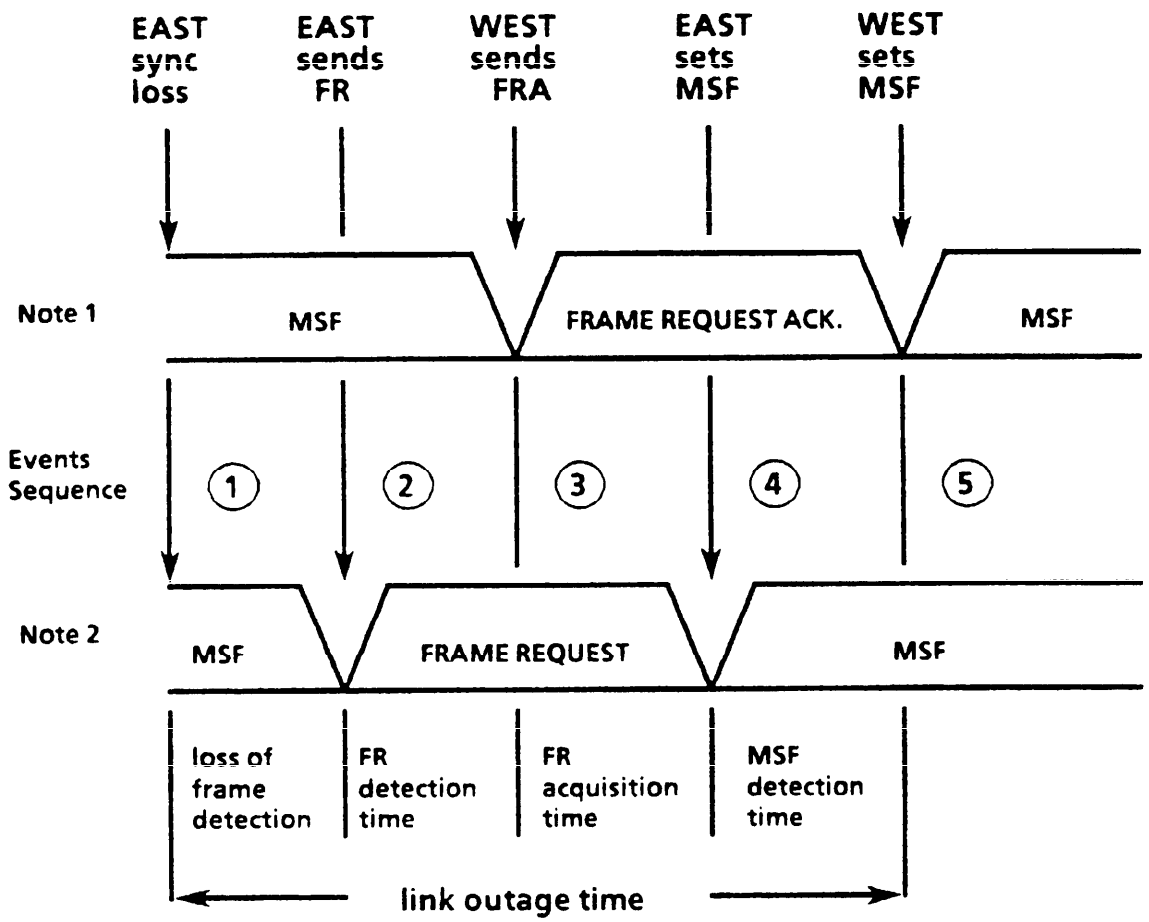
a. An equipment, designated EAST, which has lost major frame synchronization due to the inability to recognize a valid in-sync frame pattern (1010), shall: (1) discontinue the current MSF in the Go direction (i.e., the MSF outgoing and toward the distant equipment), (2) initiate the Frame Request signal in the Go direction, and (3) enable search of the return bit stream for either a Frame Request signal or a Frame Request Acknowledge signal.

b. The distant equipment, designated WEST, shall monitor the return bit stream for the Frame Request signal. Recognition of the signal shall cause WEST to replace the current MSF in the Go direction with the Frame Request Acknowledge signal.

c. EAST, upon recognition of the Frame Request Acknowledge signal, shall: (1) terminate the Frame Request signal, and (2) set the MSF in effect.

d. WEST shall monitor the return bit stream for the MSF in effect. Recognition of this format shall cause WEST to: (1) terminate the Frame Request Acknowledge signal, and (2) set the MSF in effect.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.



NOTES:

1. EAST - GO, WEST - RETURN
2. WEST - GO, EAST - RETURN

FIGURE 7. Resynchronization time-line diagram.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

5.2 Combined digital orderwire. The combined digital orderwire (CDOW) shall consist of a 16-kb/s digital voice orderwire (DVOW) multiplexed with a 2-kb/s telemetry digital orderwire (DOW) into a 32-kb/s bit stream. The CDOW signal shall not be used when the analog maintenance orderwire is used. The combined bit stream shall alternate between voice and telemetry bits, resulting in redundant transmission of the telemetry. Before combination with the voice, the telemetry shall be EXCLUSIVE OR'd with an 8-kHz clock signal. This clock signal is phased at the transmitter so that it is set LOW (digital 0) at transitions of the telemetry data (see figure 8). When the secure voice orderwire or the telemetry channel is idle or disconnected, the input state shall be all "1"s. (DIFFERENT FOR ENTIRE SECTION No equivalent paragraphs in STANAG)

5.2.1 Electrical characteristics. The CDOW shall be transmitted unmodulated along with traffic line data for any traffic line bit rate specified herein.

5.2.1.1 Transmitter.

5.2.1.1.1 Waveform. The CDOW waveform for transmission shall be Non-Return-to-Zero (NRZ).

5.2.1.1.2 Bit rate. The CDOW bit rate shall be 32 kb/s.

5.2.1.1.3 Level. The amplitude of the transmit CDOW signal shall be a postive 0.5 volts plus or minus 10 percent for a logical 1, and a negative 0.5 volts plus or minus 10 percent for a logical 0 as measured into a 58-ohm dummy load.

5.2.1.1.4 Output filtering. The CDOW transmit signal shall be filtered by the equivalent of a 4-pole low-pass Butterworth filter with a 3-dB point at 20 kHz.

5.2.1.2 Receiver. The receiver shall be capable of receiving and processing a CDOW signal transmitted through the CX-11230 coaxial cable for the range of distances specified in 4.2.3.

5.2.2 Format characteristics.

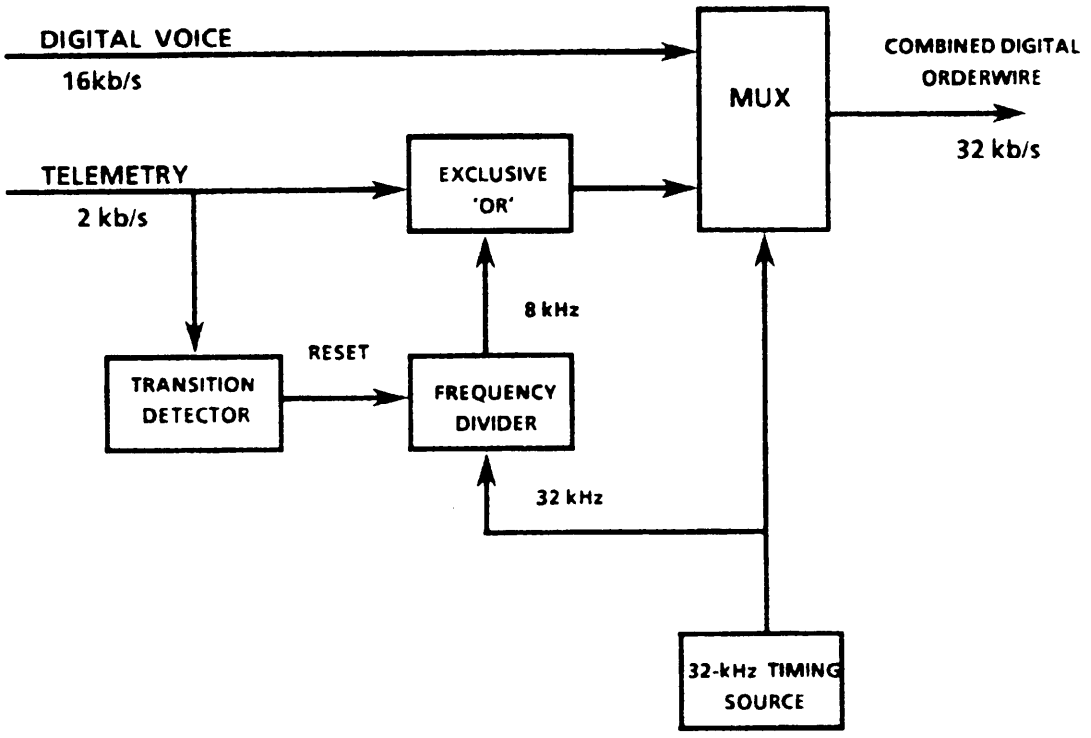
5.2.2.1 Digital voice orderwire. The digital voice orderwire (DVOW) shall be a 16-kb/s (NRZ) digital signal.

5.2.2.1.1 Digital voice orderwire signaling. A set of sixteen 8-bit codes shall be used as ring addresses. These are listed in table VI.

5.2.2.2. Digital orderwire. The digital orderwire (DOW) shall be a 2-kb/s NRZ digital signal.

5.2.2.2.1 Telemetry-combining function. The telemetry-combining function shall multiplex up to thirteen 150-b/s digital data streams to obtain one 2-kb/s digital orderwire signal.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.



Explanations:

TRANSITION DETECTOR	-	generates a RESET command for each transition of the telemetry data to synchronize the FREQUENCY DIVIDER.
FREQUENCY DIVIDER	-	divides the 32-kHz clock down to 8 kHz.
32-kHz TIMING SOURCE	-	provides the 32-kHz clock.
EXCLUSIVE 'OR'	-	generates an 8-kb/s sampling of the 2-kb/s telemetry signal.
MUX	-	multiplexes the digital voice signal with the sampled telemetry signal to generate the 32-kb/s combined digital orderwire.

FIGURE 8. Formation of combined digital orderwire.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

TABLE VI. Party-line signaling codes.

PARTY	CODE	PARTY	CODE
1	10000001	9	10110001
2	10000010	10	10101001
3	10000100	11	10110100
4	10001000	12	10110010
5	10000111	13	10011111
6	10001110	14	10101111
7	10011100	15	10110111
8	10111000	16	10111011

5.2.2.2.1.1 Multiplex format. The telemetry-combining function shall generate a combined side digital stream at a 2.0-kb/s bit rate. The frame format for 2.0-kb/s operating rates shall be comprised of a serial bit stream that is 400 bits in length. The frame is:

$$S_1S_2C_1\text{---}C_{13}F_1F_2F_3C_1\text{---}C_{13}F_4F_5F_6C_1\text{---}C_{13}F_7F_8F_9$$

where S1, S2 = 8-bit sync byte (ASCII character sync)

Cn = 8-bit ASCII character for source n, where n = 1 to 13 lower bit rate digital data streams. Unused character inputs (when data from less than 13 sources is represented in the data streams) are coded as 11111111. When teletypes or other character asynchronous sources are used, the idle pattern (11111111) may be inserted by the multiplexing process when two adjacent characters become sufficiently separated. Each n identifies a character (information) stream from a separate generating source, where n = 1 to 13.

Fm = idle character 11111111, generated by the telemetry combiner in the multiplexing process to obtain the frame format, where m is an integer.

5.2.2.2.2 Data channel multiplexer function. The data channel multiplexer function shall multiplex up to seven 2-kb/s digital data subchannels to obtain one 16-k/bs channel, or up to seven 4-kb/s digital data subchannels into one 32-kb/s channel.

5.2.2.2.2.1 Multiplex format. The data channel multiplexer function frame for the 16/32-kb/s data stream shall consist of 32 bits and shall be divided into four 8-bit subframes. Each subframe shall consist of one overhead frame bit (O_h) followed by seven data bits with one bit selected sequentially from each of the seven subchannels (S_n). (Unused or idle subchannels will output a continuous ONEs signal.) The frame synchronization pattern shall consist of one bit in the fourth subframe. The 32-bit frame shall be represented as:

$$O_1S_1\text{---}S_7,\ O_2S_1\text{---}S_7,\ 0_3S_1\text{---}S_7,\ 0_4S_1\text{---}S_7,\ O_1S_1\text{---}$$

Source: https://assist.dla.mil -- Downloaded: 2026l0-26T07:28Z
Check the source to verify that this is the currentversion before use.

Where $O_1 = O_2 = O_3 =$ overhead bit set to ZERO.

S_1 through $S_7 =$ output bit for subchannel, 1 through subchannel 7, set to ONE for unused or idle subchannel(s).

$O_4 =$ overhead bit set to ONE

Thus, the frame format is:

OS_1---S_7, OS_1---S_7, OS_1---S_7, $1S_1$---S_7

5.3 Analog maintenance orderwire. The analog maintenance orderwire can be used in lieu of the combined digital orderwire (CDOW) and shall have the following characteristics. (DIFFERENT FOR ENTIRE SECTION No equivalent paragraphs in STANAG)

5.3.1 Electrical characteristics. The analog maintenance orderwire shall be transmitted unmodulated along with the traffic line signal.

5.3.1.1 Transmitter.

5.3.1.1.1 Waveform. The analog maintenance orderwire signal shall be transmitted unmodulated.

5.3.1.1.2 Signaling. The analog orderwire shall employ an in-band tone-signaling frequency of 1600 Hz.

5.3.1.1.3 Level. The power level of the transmit signal shall be + 5 dBm plus or minus no greater than 1 dB with a 0.442 volts root-mean-square (Vrms) 1000 Hz test-tone input.

5.3.1.2 Receiver. The receiver shall be capable of receiving and processing an analog maintenance orderwire signal transmitted through the CX-11230 coaxial cable for the lengths specified in 4.2.3.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

6. NOTES

6.1 Subject term (key word) listing.

Cable Interface
Coaxial Cable
Digital Transmission Group
Multiplex Signaling Format
Orderwire
Time-Division Multiplexing

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

THIS PAGE INTENTIONALLY LEFT BLANK

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

APPENDIX A

ACRONYMS AND ABBREVIATIONS

This appendix provides definitions of acronyms and abbreviations used in this standard. This appendix is not a mandatory part of this standard.

ACK	Acknowledge
ASCII	American Standard Code for Information Interchange
AVOW	Analog Voice Orderwire
BER	Bit Error Ratio
CCITT	Consultative Committee for International Telegraph and Telephone
CDOW	Combined Digital Orderwire
CVSD	Continuously variable slope delta (modulation)
dB	Decibel
dBm	Decibels referenced to 1 milliwatt
DCA	Defense Communications Agency
DCAC	Defense Communications Agency Circular
DCS	Defense Communications System
DO	Design Objective
DoD	Department of Defense
DoDD	Department of Defense Directive
DoDISS	Department of Defense Index of Specifications and Standards
DOW	Digital Orderwire
DTG	Digital Transmission Group
DVOW	Digital Voice Orderwire
FR	Frame Request
FRA	Frame Request Acknowledge
Hz	Hertz
ISDN	Integrated Services Digital Network
JCS	Joint Chiefs of Staff
kb/s	Kilobits per second
kHz	Kilohertz
km	Kilometer(s)
MHz	Megahertz
MIL-STD	Military Standard
msec	Millisecond(s)
MSF	Multiplex Signal Format
NATO	North Atlantic Treaty Organization
NMCS	National Military Command System
NRZ	Non-Return-to-Zero
OASD	Office of the Assistant Secretary of Defense
p-p	Peak-to-peak
ppm	parts per million
rms	Root-mean-square
STANAG	Standardization Agreement (NATO)
TDM	Time-division Multiplexing
V	volt(s)
Vp-p	volts peak-to-peak
Vrms	volts root-mean-square

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

THIS PAGE INTENTIONALLY LEFT BLANK

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

APPENDIX B

TYPICAL APPLICATION OF TYPE 3 AND TYPE 4 GROUPS

Appendix B contains application information in support of MIL-STD-188-202 and is a non-mandatory part of this document.

Figure 9 illustrates a Type 3 Group typical application between a nodal switch and a remote radio assemblage.

Figure 10 illustrates a Type 4 typical application between a large circuit switch and a collocated nodal control element.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

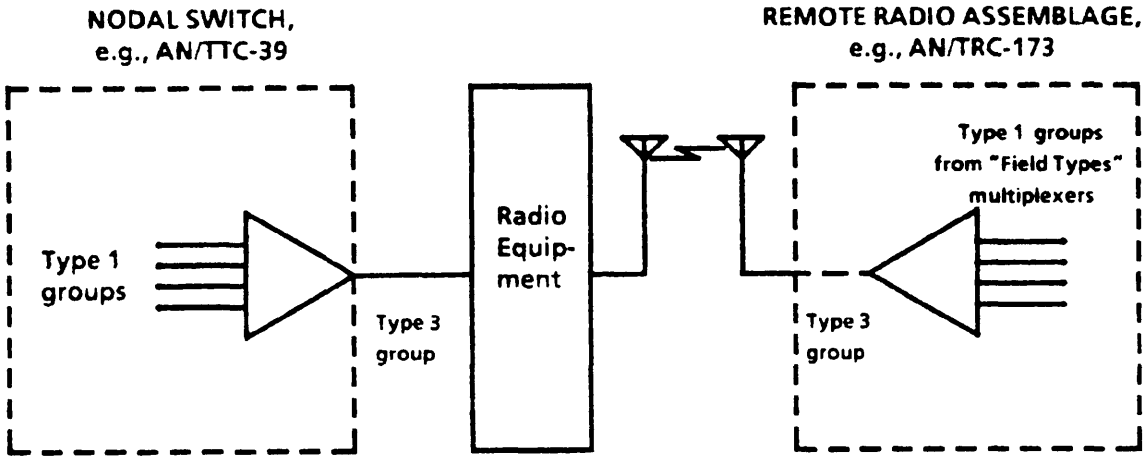


FIGURE 9. Type 3 group, typical application.

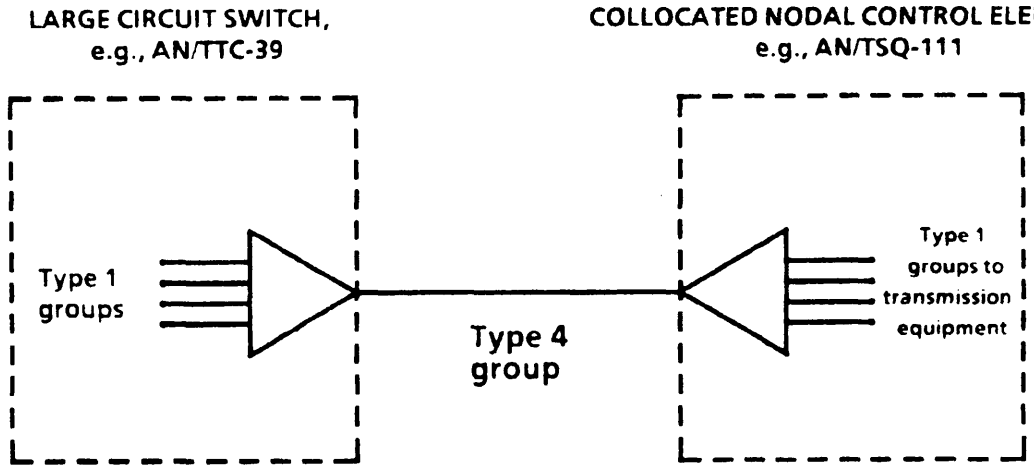


FIGURE 10. Type 4 group, typical application.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

APPENDIX C

MILITARY STANDARD AND NATO STANAG COMPATIBILITY

This appendix is provided for information purposes only and is a non-mandatory part of this document.

Table VII is a comparision matrix of key mechanical, electrical, and multiplex format characteristics as stated in this MIL-STD-188-202 and the following NATO STANAG documents:

. STANAG 4206	The NATO Multi-channel Tactical Digital Gateway - System Standards.
. STANAG 4207	The NATO Multi-channel Tactical Digital Gateway - Multiplex Group Framing Standards.
. STANAG 4209	The NATO Multi-channel Tactical Digital Gateway - Standards for Analogue to Digital Conversion of Speech Signals.
. STANAG 4210	The NATO Multi-channel Tactical Digital Gateway - Cable Link Standards.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

Table VII. Comparison of MIL-STD-188-202 and STANAGs 4206, 4207, 4209 and 4210

REQUIREMENT	MIL-STD-188-202	STANAGs
MECHANICAL:		
Cable Type	CX-11230	2 Balanced Pairs, Spiral Quad, 1Pair Each Direction
Cable Connector Type	UG-1870	U-176/G
Connector Convention	FEMALE - transmit	TX-transmit
Maximum length	3.2 km	2 km
ELECTRICAL:		
Group Bit Rate	128/256/512/2048 kb/s	256/512 kb/s
Modulation	Conditioned Diphase	Conditioned Diphase
Impedance	58 ohms	130 ohms
Transmit Level	3 volts p-p, +/-20%	4.6 volts p-p, +/-10%
Bit Error Rate	$< 1 \times 10E-9$	$< 1 \times 10E-09$
Phase Jitter	2.5 bits p-p 2.6 bits p-p (frequency mask)	0.4 bits p-p in 41 msec. 0.4 bits p-p in 82 msec. (time mask)
Orderwire Method	32 kb/s baseband CDOW	16 kb/s in phantom circuit of the 2 cable pairs
Voice Coding	CVSD	CVSD
MULTIPLEX FORMAT:		
Channel Bit Rate	16/32 kb/s	16/32 kb/s
Total # of Channels	8/16/32/64/128	16
# of Traffic Channels (N)	7/15/31/63/127	14
Framing Period	0.0625/0.03125 msec. (denoted subframe)	0.0625/0.03125 msec. (denoted frame)
Framing Channel	Slot 0, Subframe 0	Slot 0
Signaling Channels	Slot 0, Subframe 1-4	Slot 1
Traffic Channels	Slots 1-N, all Subframes	Slots 2-15

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

APPENDIX D

INTEGRATED SERVICES DIGITAL NETWORK (ISDN)

This appendix is provided for information purposes only and is a non-mandatory part of this document.

An OASD (C^3I) Memorandum on subject matter, dated 15 September 1986, clearly mandates that DoD Agencies must be prepared to evolve into the new technology of ISDN.

ISDN Standards have been developed by commercial carriers in the United States and by the International Telegraph and Telephone Consultative Committee (CCITT).

If the tactical community is to take advantage of the end-to-end digital service offered by the ISDN, it will be necessary to provide terminal adapters or special tactical/commercial gateways.

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:28Z
Check the source to verify that this is the current version before use.

THIS PAGE INTENTIONALLY LEFT BLANK

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:28Z
Check the source to verify that this is the current version before use.

CONCLUDING MATERIAL

Custodians:
Army -- SC
Navy -- EC
Air Force -- 90

Review activities:
Army -- CR
Navy -- MC
DoD -- DC, NS, JT

Preparing activity:
JTC3A -- JT

(Project TCTS-2020)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.

DD1426

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:28Z
Check the source to verify that this is the current version before use.